PassFrame tuition

Number Sequences & Patterns

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 10, 12

2. 25, 30

3. 9, 11

4. C — 16

5. 70, 60

6. A, C and E — each steps up by 3

7. A — add 5

8. 6

9. 35, 30

10. 28, 35

11. 30

12. 13, 16

13. B — subtract 2

14. A, C and E — each steps up by 6

15. 12

Standard

1. 32, 64

2. 25, 36

3. 15, 21

4. A, C and E — each term doubles the one before

5. C — 9

6. 19

7. 2.5, 3

8. A, B and D — the same amount is subtracted each time

9. 81

10. 8, 13

11. B — 40

12. 7, 10

13. C — 12.5

14. ×3; 162

15. 32

Standard / Hard

1. 3n + 2

2. A — n = 15 (3×15+2 = 47)

3. 3n − 1; 20th term = 59

4. 37

5. A, B, D and E — 2, 5, 10 and 17 are terms (n=1,2,3,4); 8 is not

6. 25 sticks

7. 4, 16

8. 5n + 2

9. C — no whole number of n gives 100

10. A and C — 8 and 32 are terms (n=2, n=4)

11. 16

12. 103 (n = 26)

13. ×2 then +1; 63

14. B — n = 3

15. 55

Homework — Answers

Easy / Standard

1. A, C and D — each steps up by 3

2. 30, 36

3. 14, 17

4. B — 9

6. 16, 20

7. B — add 7

8. 15

9. 42, 36

11. 36

12. 18, 22

13. B — subtract 3

14. 36, 45

5. 60, 50

10. A, C and D — each steps up by 8

15. 15

Standard

1. 80, 160

2. 36, 49

3. 28, 36

4. A, C and E — each term doubles the one before

5. B — 10

6. 11

7. 1.0, 1.2

8. A, C and E — each decreases by the same amount

9. 54

10. 21, 34

11. C — 42

12. 8, 12

13. B — 25

14. ×3; 243

15. 40

Standard / Advanced

1. 17

2. 30

3. C — n = 12 (36÷3)

4. A, C and E — 16, 25 and 36 are square numbers

5. C — the 10th term (5×10−3 = 47)

6. 11 chairs

7. 36 tiles

8. £170

9. 20cm

10. C — 62 seats

11. 46 matchsticks

12. 45 dots

13. 23cm

14. A, D and E — 100, 400 and 800 all occur (after 1, 3 and 4 hours)

15. 3, 5, 9, 17, 33